HACK AMONG US

SMART CONTRACT TERMS OF SERVICE HUMANIZER

TRACK: FINTECH FRONTIER

TEAM: CREWMATES

TEAM MEMBERS:

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Problem Statement

What are the challenges?

- Most people **sign digital finance contracts** (ToS) **without reading** or **understanding** them
- **Hidden fees** and **dangerous clauses** go unnoticed everyday by users
- Blockchain and financial agreements are **too complex for non experts**
- **No affordable tool exists** to check contracts safely for regular users
- Signing bad contracts unknowingly can lead to **serious financial loss**

What is the Problem Impact ?

- Over **$14B lost** to crypto scams and bad contracts **in 2023**
- **33% Decentralized Finance** contracts has at least **one clause that harms** the user
- Most **people blindly trust**, with **no way to verify** what they're signing
- **Risk grows** as more people enter Web3 and digital finance
- Readable contracts are **basic financial safety need**

Problem Solution

How the Problem is solved?

- Built a tool where users **upload any contract** (PDF) and **get a risk report**
- **AI scans and matches it** against a dataset of known bad clauses
- **Flags risky language** with a plain English explanation and **risk score**
- **Highlights** exactly **which parts are dangerous** and why
- No legal or technical knowledge needed **built for everyone**

Key Features of solution model

- **Rates the overall contract** on a risk score from 0 to 100
- Scans for **hidden fees, rug-pulls** and **other dangerous patterns**
- Integrated **chat-bot for detailed information** of contracts
- **Three risk checks run** automatically the moment a file is uploaded
- Results displayed in a clean, simple dashboard fast and easy to read

Technical Stack

Frontend

ReactJS
for smooth lightwight UI

Tailwind-CSS
for consistent and responsive UI

API

HuggingFace Model
for generating Embeddings (text vectors from docs)

Gemini API
for Information Retrival and chatbot

Backend

NodeJS
for JS runtime Environment in server

ExpressJS
for Handling Web-requests

Flask
for connecting the RAG pipeline to the app as microservice

Authentication

JsonWebToken
for validating user without storing session data in server

Rag Pipeline

LangChain
for building the RAG pipeline

PyMuPDF
for reading and extracting text from PDF/Docs

Database

MongoDB
for storing user information

ChromaDB
as a vector database for storing the text vectors

Implementation Plan

How the solution is implemented (with RAG)

- **Step 1:** Built a dataset of known bad contract clauses and scam patterns
- **Step 2:** Connected it to a search system that finds similar patterns in new contracts
- **Step 3:** Pass results to Gemini (AI model) that writes a structured risk report in JSON
- **Step 4:** Built a simple frontend that turns the JSON into a readable dashboard
- **Step 5:** Test against real world honeypot and rug-pull contracts

Impacts

Benefits

Impact on Users

- Check any contract before signing no expertise needed
- Instantly understand risky clauses in plain English
- Avoid hidden fees, scams, and financial loss

Benefit For Users

- Plain English results no expertise needed
- Full scan in seconds, faster than manual review
- Based on real scam data not guesswork

Impact on Developers

- Pre-deployment scan for bugs and bad patterns
- Catch vulnerabilities early, before they cost money
- Faster and cheaper than a manual security audit

Benefit For Developers

- Early bug detection saves deployment costs
- Easy to integrate via structured JSON output
- Reduces need for expensive third-party audits

Impact on Organizations & Institutions

- Teaches contract safety through real-world examples
- Quick preliminary screening — no legal team needed
- Scales across departments with zero legal overhead

Benefit For Organizations & Institutions

- One tool scales across teams and departments
- Standardizes and lowers contract review risk
- Makes safety a built-in, trust-building feature

References

- **Retrieval-Augmented Generation (RAG) :** https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-retrieval-augmented-generation-rag
- **Chainalysis Crypto Crime Report :** https://go.chainalysis.com/crypto-crime-report.html ($14B+ crypto scams)
- **SWC Registry :** https://swcregistry.io (knowledge base)
- **RAG Paper (Meta AI) :** https://arxiv.org/abs/2005.11401
- **Consumer Financial Protection Bureau (CFPB) :** https://www.consumerfinance.gov (Predatory Clauses & Financial Inclusion)

THANK YOU

We believe this assistant represents the future of enterprise support systems.